
Milestone 4 — Deployment & User Interface

AI TraceFinder: Scanner Identification System

1. Overview

Milestone 4 focuses on deploying the trained **CNN-based AI TraceFinder model** as a real-time web application using **Streamlit**.

The objective of this milestone is to convert the deep learning model into a usable forensic tool that allows users to upload scanned document images and obtain:

- Predicted Scanner
- Predicted DPI
- Predicted Source Type (Official / Wikipedia / FlatField)
- Prediction Confidence Score
- Prediction logging and CSV export

This milestone transforms the research model into a complete end-to-end forensic system.

2. Objective

The main objectives of Milestone 4 are:

- Deploy the trained CNN model using Streamlit
 - Provide a user-friendly interface for scanner identification
 - Enable real-time inference on uploaded document scans
 - Display prediction confidence
 - Maintain logs of predictions
 - Allow exporting results as CSV
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3. System Architecture

The deployed system follows the architecture below:

User Uploads Image

- Image Preprocessing (Resize, Normalize)
 - CNN Feature Extraction and Classification
 - Prediction and Confidence Score
 - Display Results on UI
 - Save Logs to CSV
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5. Application Features

5.1 User Interface

The Streamlit interface provides:

- Image upload functionality
 - Display of uploaded scanned image
 - Real-time prediction button
 - Output display panel
 - Confidence score visualization
 - Prediction log viewer
 - CSV export functionality
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5.2 Prediction Logging

Each prediction is stored with the following details:

- Image filename
- Predicted scanner
- Predicted DPI
- Predicted source
- Confidence score
- Timestamp

Logs are stored in:

logs/prediction_logs.csv

5.3 CSV Export

Users can download the prediction logs for:

- Dataset analysis
 - Performance monitoring
 - Documentation
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6. Application Workflow

1. User uploads a scanned document image
 2. Image is resized and normalized
 3. CNN processes the image
 4. Model predicts scanner, DPI and source
 5. Confidence score is computed
 6. Results are displayed on UI
 7. Prediction is saved in logs
 8. User can download results as CSV
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7. Deployment

Local Deployment

```
streamlit run app.py
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8. Conclusion

Milestone 4 completes the AI TraceFinder project by deploying a CNN-based forensic scanner identification model through a Streamlit web interface. The system enables real-time prediction of scanner source, DPI, and document origin with high confidence and reliability.

This milestone demonstrates the successful translation of deep learning research into a practical forensic application.
